# Race to Idle: Efficiency Cores Do Not Save Energy for LLM Decode on x86 Heterogeneous SoCs

Anonymous Author(s)

***Abstract*—Heterogeneous consumer SoCs expose core islands whose names—*performance*, *efficiency*, *low-power efficiency*—carry an implicit scheduling recommendation. We test that recommendation for large language model decode and find it inverted. Using Intel RAPL counters rather than estimates, we take 240 measurements across twelve configurations of precision and core placement on an Intel Core Ultra 7 155H, timing prefill and decode separately. At fixed precision and byte-identical output, the low-power island costs $2.306\times$ [2.301, 2.314] the energy per generated token of the performance island. The mechanism is directly observable: gross package power is 14.00 W for every fp32 and bf16 configuration, across islands whose throughput spans 7.0 to 16.1 tok/s, against a 40 W platform limit. With power constant, energy per token reduces to duration, and the measured energy ratios match inverse throughput to within 1.1%. Decode is memory-bandwidth-bound, so the island issuing the loads barely affects the power drawn, and the correct policy is to race to idle. The identity fails only for int8, the one precision whose power is not flat, which we take as confirmation rather than counterexample. We further find that the bf16 energy advantage disappears entirely on the low-power island, so precision and placement cannot be chosen independently; that per-tensor int8 saves 27% energy while raising perplexity by 785%; and that the effect strengthens at smaller model scale, reaching $3.712\times$ on a 135 M model. Our result is the opposite of one reported for ARM mobile, and we identify the single measurement that distinguishes the two cases.**

***Index Terms*—Energy measurement, RAPL, large language models, on-device inference, heterogeneous multicore, quantization, sustainable computing.**

## I. Introduction

LANGUAGE model inference is moving onto the device. The hardware it lands on is not the homogeneous server CPU for which most energy-efficiency intuitions were formed, but a heterogeneous SoC with several core islands of differing capability: on the Intel Core Ultra 7 155H studied here, six performance (P) cores with SMT, eight efficiency (E) cores, and two low-power efficiency (LP-E) cores on a separate power domain.

A practitioner deploying a model on such a part must choose a numeric precision and a core placement. Both choices are usually made by guesswork, or by throughput, because the data needed to make them properly does not exist. Energy figures for LLM inference are overwhelmingly reported for datacenter GPUs [1], [2], and the figures that do exist for edge parts are typically derived from thermal design power or vendor power models rather than measured.

The island names themselves constitute advice. "Efficiency core" asserts that work placed there costs less energy. That assertion was formed on mobile workloads which are bursty and latency-tolerant, and it is not obvious that it survives transfer to autoregressive decode, which is sustained and memory-bound. This paper tests it.

We measure rather than estimate. Intel's Running Average Power Limit (RAPL) interface exposes energy counters that have been validated against external instrumentation [3], and we read them directly, subtracting a measured idle baseline and reporting both gross and net energy. We separate prefill from decode, because they have opposite cost structure and a single wall-clock figure per request conceals the quantity an energy study exists to measure. We record the KV cache size at every decode step, because on long-context work it, rather than the weights, dominates memory traffic.

Our central finding contradicts the island naming. At fixed precision, with byte-identical generated output across placements, the low-power island costs $2.306\times$ [2.301, 2.314] the energy per generated token of the performance island. The reason is visible in the power column rather than inferred from the energy column: gross package power across islands differs by under 0.1%, while throughput differs by $2.3\times$. The low-power island does not draw low power on this workload. It draws approximately the same power for considerably longer, and the measured energy ratio matches inverse throughput to within 1.1%—the classical race-to-idle result [4], [5], re-derived for a workload class in which it was not obvious it would hold.

That invariance is not an accident of this part, and it is not a power cap: the package draws 14.00 W against a long-term RAPL limit of 40 W. It follows from decode being memory-bandwidth-bound (Section II-B). When the bottleneck is the memory system, the identity of the cores issuing the requests has little influence on the power drawn, so moving work to a nominally lower-power island changes duration without changing power—which is the worst possible trade.

This also sets the boundary of the claim, which we state in full in Section VI. A platform whose low-power island does achieve a materially lower operating power under this workload would admit the trade, and indeed one has been reported on ARM mobile [6]. We address that contradiction directly in Section VII-A; the short version is that the two results describe different hardware, and that a single measurement—whether package power varies across islands—distinguishes the cases.

Author and affiliation details are withheld for double-anonymized review and are supplied on the separate title page.

The artifact—measurement harness, complete dataset and every discarded run—is archived under a permanent DOI, withheld here because the repository identifies the author. It is supplied on the title page and will be stated in the camera-ready version.

### A. Contributions

- A direct test of the compute-bound/memory-bound premise the analysis rests on: simultaneous multithreading makes prefill $1.193\times$ faster and decode $0.936\times$ slower, measured on the same hardware in the same runs (Section IV-H). The supporting harness separates the two phases, records KV cache growth per step, subtracts a verified idle baseline, and degrades to timing-only measurement rather than silently estimating energy when counters are unreadable. It is released with a unit test suite covering the logic that would corrupt results without failing loudly (Section III).
- The race-to-idle result for LLM decode on a heterogeneous consumer SoC, with the mechanism isolated by measuring power and duration separately rather than reporting their product (Section IV-A).
- An energy accounting of per-tensor dynamic int8 quantization: it is genuinely cheaper, by 27%, and unusable, raising perplexity by 785% and eliminating needle retrieval; and it leaves the KV cache untouched—so the operative variable is quantization *granularity*, not bit width (Section IV-D).
- Two findings with direct scheduling consequences: the bf16 energy advantage is island-dependent and vanishes on the low-power island, so precision and placement interact; and core placement perturbs the numerics of quantized inference measurably without changing greedy output (Sections IV-C and IV-G).
- A paired experimental design and the statistical treatment it enables, including a negative result: one of the comparisons frequently made in this area cannot be resolved by data of this kind, and we say so rather than report the point estimate (Section III-C).

## II. Background

### A. RAPL

RAPL exposes per-domain energy counters as monotonically increasing microjoule totals. Three properties shape any harness built on it.

First, the counters *wrap* at a domain-specific maximum, so a naive end-minus-start subtraction goes negative once per wrap period; correct accounting requires the wrap bound, and measurement windows must stay well inside it.

Second, the counters are root-readable only on current kernels. This is a mitigation for a class of software power side-channel attacks: PLATYPUS recovered AES-NI keys from SGX enclaves and broke KASLR using nothing but unprivileged RAPL access [7], and related work extracted further secrets through RAPL-induced channels [8]. Any reproduction of this work therefore requires an explicit, deliberate relaxation of permissions, which we document rather than assume.

Third, on integrated parts the package domain includes the iGPU. CPU and iGPU energy cannot be fully separated within a single run, which bounds what this methodology can claim about engines other than the CPU.

RAPL's accuracy for this purpose has been characterised independently: Khan et al. [3] report that RAPL readings predict full-system power closely, that measurement overhead is negligible, and that the counters update on microsecond timescales on recent microarchitectures.

### B. Prefill and decode

Autoregressive generation has two phases with opposite cost structure. Prefill processes the whole prompt in a single forward pass: the weights are reused across many tokens, arithmetic intensity is high, and the phase sits at or beyond the ridge point of the roofline. Decode emits one token per forward pass. Each pass is a sequence of matrix–vector products in which every weight is read once and used once, giving an arithmetic intensity of order one FLOP per byte and placing decode far down the memory-bound slope of the roofline [9]. The consequence is well known in serving systems, where memory rather than arithmetic is the resource worth managing [10].

This distinction is the reason we time the two phases separately, and it is also what makes the central result of this paper predictable in hindsight. If decode is bounded by memory bandwidth, then the hardware doing the waiting matters less than the memory system being waited on. A core island that issues the same stream of requests more slowly does not thereby draw proportionally less power, because the power is not going into the cores.

A concrete figure for the platform studied here: a 0.5 B-parameter model in fp32 requires roughly 2 GB of weight traffic per generated token, so sustained decode at order ten tokens per second implies order $20\,\mathrm{GB\,s^{-1}}$ of DRAM traffic. Under that load the package power budget is dominated by the memory controller, the interconnect and static draw rather than by core execution, and the cores spend most cycles stalled.

### C. Heterogeneous core islands

Heterogeneous CPUs expose islands intended for different operating points, but the design intent behind the labels differs sharply between ecosystems, and this turns out to matter for interpreting energy results.

ARM big.LITTLE pairs out-of-order performance cores with in-order cores designed to operate at a small fraction of the performance cores' power. The intuition that migrating work to the small cores saves energy was formed on that hardware and on workloads—bursty, latency-tolerant, often interactive—for which it holds.

Intel's heterogeneous parts divide differently. The E-cores are throughput-oriented rather than low-power-oriented: they deliver area-efficient multithreaded throughput, and they are not positioned as a low-power operating point. Only the LP-E island, two cores on a separate power domain, is positioned that way. "Efficiency core" therefore names two materially different things in the two ecosystems, which is a trap for anyone carrying an energy intuition across. Section VII-A returns to this when reconciling our result with the opposite finding reported on ARM mobile [6].

The energy behaviour of these designs has been studied extensively on ARM parts [11], [12], [13], and the trade is real there; Section VII-A examines why it does not follow for the platform measured here.

### D. Quantization

Post-training quantization of LLM weights is well studied, and the literature is clear that *how* scales are assigned matters at least as much as how many bits remain. LLM.int8() isolates outlier features and handles them at higher precision [14]; GPTQ performs second-order-informed per-layer rounding [15]; AWQ scales channels by activation magnitude [16]. All three preserve quality at bit widths where naive schemes fail. The scheme we evaluate here—PyTorch dynamic quantization, which assigns one scale per tensor—is the naive end of that spectrum, and our results should be read as a measurement of that granularity rather than of 8-bit arithmetic in general.

## III. Methodology

### A. Platform

All measurements were taken on a single Intel Core Ultra 7 155H (Meteor Lake) with 15 GB of memory, running Linux 6.8.0-138, Python 3.10.12, PyTorch 2.7.1 and Transformers 4.45.0. The part exposes three core islands:

| Island | Physical | Logical | Max GHz |
|---|---|---|---|
| P (performance) | 6 | 12 | 4.8 |
| E (efficiency) | 8 | 8 | 3.8 |
| LP-E (low-power eff.) | 2 | 2 | 2.5 |

Island membership is derived at runtime from `cpuinfo_max_freq` rather than from a hardcoded CPU model, so the harness degrades sensibly on other parts: a homogeneous CPU yields a single tier. Meteor Lake reports two P-core turbo bins (4.8 and 4.5 GHz) which are the same core type; tiers within 10% of one another are merged rather than reported as distinct islands.

### B. Configuration matrix

We cross three numeric precisions (fp32, bf16, per-tensor dynamic int8) with four core placements (P-1t: six P-cores, one thread each; P-smt: all twelve P threads; E: eight E-cores; LP-E: two low-power cores), giving twelve configurations. Threads are pinned with explicit affinity masks, and the mask is restored after each configuration so placements cannot leak into one another.

All measurements use **batch size one**. This is the on-device regime, in which requests arrive singly and there is nothing to batch with, and it is the regime in which decode is most firmly memory-bound. Batching raises arithmetic intensity by amortising each weight read across several sequences, so a batched server may not exhibit the power invariance reported here; we return to this in Section VI.

### C. Paired design

Each configuration is measured on four fixed prompts, with five true repeats per prompt: twenty measurements per configuration, 240 in total. The distinction between the two axes matters. Repeats are exchangeable measurements of one quantity and bound measurement noise; prompts are not exchangeable and differ systematically in length and content.

Because every configuration sees the same four prompts, comparisons are formed *within* a prompt and then aggregated, which removes prompt-to-prompt variation entirely. This is not a cosmetic refinement. On this workload prompt variance is comparable in magnitude to the effects of interest, so the unpaired comparison buries real effects inside the prompt spread; we quantify the difference in Section IV.

We report the geometric mean of per-prompt ratios with a percentile bootstrap 95% confidence interval over prompts, and we state the direction of an effect only when that interval excludes unity; Fig. 2 collects every comparison in the study on a single axis.

The resampling unit is the prompt, and there are four of them. Four values cannot support asymptotic coverage, and we do not claim it: these intervals are *descriptive ranges* summarising how closely the per-prompt ratios agree, not inferential confidence intervals in the usual sense, and we write them as brackets rather than as $\pm$ to signal that. We prefer that to the alternative: with twenty measurements per configuration we could resample individual trials and obtain far narrower intervals, but those would understate the uncertainty that matters, which is whether an effect holds across inputs rather than across repetitions of one input. Where the per-prompt ratios disagree in sign we report that fact instead of an interval. We also report whether the ratio has the same sign in every prompt, which distinguishes a consistent effect from one carried by a single input.

### D. Measurement discipline

A laptop is a hostile place to measure energy, and treating it otherwise is how results of this kind are correctly dismissed. The harness enforces five controls.

*Idle baseline.* Measured once per run on a quiet machine before any load and subtracted from every reading. At these batch sizes idle draw is a large fraction of the total, so gross joules would largely measure the machine being switched on. Both gross and net are recorded.

*Warmup to steady state.* Each configuration is warmed by generating at the full measurement length, repeatedly, until two consecutive warmups agree within 5%; two to five rounds were required here. All warmups are discarded. A single short warmup is not sufficient and we report the evidence: with one 8-token warmup, decode throughput was still climbing when the measured trials began, producing a $+20.5\%$ drift *within* one configuration at 256 tokens. Because the drift size differed by configuration—$+20.5\%$ for fp32/P-1t against $+1.8\%$ for bf16/P-1t—it did not cancel when configurations were compared, and it inverted the bf16-versus-fp32 result. That dataset is released alongside this one. Each row records whether its warmup converged, so a configuration that never settled is visible in the data rather than silently treated as warm.

*Repeats and robust statistics.* Timing distributions here are right-skewed—background activity adds a long tail and nothing pulls the other way—so we report medians rather than means.

*Fixed inputs.* Identical prompts, seed and perplexity corpus for every configuration. Perplexity is only ever compared between precisions of the same model on the same corpus with the same tokenizer.

*Unclamped net energy.* Net energy is recorded without flooring at zero. A negative value means the baseline exceeded the measurement, which is a measurement error worth seeing rather than hiding.

### E. What the harness refuses to do

When RAPL counters are unreadable the harness records energy as null and both the analysis and the generated report state plainly that the cost axis fell back to seconds per token. Nothing is estimated from TDP and then presented as measured. Energy availability is resolved once per run rather than per measurement, so a permissions change mid-run cannot produce a file in which some configurations carry joules and others do not—which would make any frontier computed over it compare incomparable quantities.

### F. Quality measurement

We report perplexity over a fixed public-domain corpus and a needle-in-context retrieval probe at several depths. This is a deliberately thin quality measure and we treat it as a tripwire rather than an evaluation: it is adequate to detect a collapse, not to rank configurations that are all working.

## IV. Results

Table I reports all twelve configurations. Intervals throughout are percentile bootstrap intervals; ratios are paired per prompt as described in Section III-C.

We report *gross* package power alongside net energy and treat it as the primary evidence, because it is the quantity we actually measure. Net energy subtracts an idle baseline of 11.23 W from a gross draw of 14.00 W, so about 80% of each reading is a subtraction and the absolute net joules inherit the baseline's uncertainty. The comparisons in this section do not: where two configurations draw equal power, their net energy ratio equals their duration ratio whatever baseline is removed, so every ratio reported below is baseline-independent while the absolute figures in Table I are not. Readers who distrust the baseline can read the gross-power column and the throughput column and reconstruct every claim.

### A. Core placement: energy per token is duration

At fixed precision and byte-identical output, joules per token ordered by placement, as paired ratios against the performance cores:

| bf16 placement | J/token | ratio vs. P-1t |
|---|---|---|
| P-1t | 0.1758 | — |
| P-smt | 0.1878 | 1.073 [1.066, 1.081] |
| E | 0.1987 | 1.148 [1.117, 1.189] |
| LP-E | 0.4053 | **2.306 [2.301, 2.314]** |

The low-power island costs $2.306\times$ the energy per generated token of the performance island. The decisive observation, however, is in the power column of Table I: **gross package power is 14.00 W for every fp32 and bf16 configuration**, across islands whose throughput spans 7.0 to 16.1 tok/s. The medians are 14.00, 14.00, 14.01, 14.00, 14.00, 14.00, 14.00 and 14.01 W. This is not a power cap; the long-term RAPL limit on this part is 40 W.

Flatness of this degree invites a measurement objection: some Intel parts derive package power from an internal model rather than from direct sensing, and a saturating model would produce exactly this appearance. The data refutes that here. The int8 configurations, measured by the same counter on the same part in the same runs, span 13.96 to 15.92 W (Section IV-B). The instrument resolves power differences on this platform; fp32 and bf16 simply do not produce any.

With power constant, energy per token reduces to duration, and that is a prediction rather than a description. Fig. 1 shows the two axes together, and Table II tests the prediction by comparing each configuration's measured energy ratio against the inverse of its independently measured throughput ratio.

There is a name for what the power column shows. Barroso and Hölzle [17] argued that systems should be *energy-proportional*—that power should track utilisation—and observed that real machines are not, drawing substantial power at low load. Our measurement is that instance taken to its limit across a different axis: moving the same work between islands changes utilisation of each island completely and package power not at all. Where a system is perfectly non-proportional in this sense, energy reduces to time, and the only available lever is to finish sooner.

Two further consequences follow and are worth separating. The ordering is robust: it rests on a $2.3\times$ duration difference against a power difference of under $0.1\%$, so no plausible baseline error inverts it. And the framing "efficiency cores save energy" fails here not because the saving is small, but because there is no power saving available to trade in the first place.

### B. The identity fails exactly where the mechanism predicts

int8 is the one precision whose power is not flat: it spans 13.96–15.92 W, median 14.81 W. It is also the one precision for which the identity breaks, by up to 27.5% against a worst case of 1.10% for fp32 and bf16 (Table II). Quantization moves work off the memory system and back into the cores; power ceases to be flat, and duration ceases to be the whole story.

We regard this as the strongest evidence for the explanation in Section IV-A. A flat-power observation alone is compatible with several accounts. An account that also predicts *where* flatness should fail, and is then confirmed by the one precision that departs from it, is a mechanism.

A second engine supports the same conclusion. Running GGUF models through llama.cpp—a different runtime with its own kernels, written in C++ rather than dispatched through PyTorch—reproduces the invariance: gross package power is 14.01 W for `Q4_K_M` and 13.99 W for `Q8_0`, spanning 16.4 to 31.7 tok/s across the two islands, against the 14.00 W measured under PyTorch. The duration identity holds there too, to within $1.1\%$ ($1.660\times$ energy against $1.678$ inverse throughput for

TABLE I
ALL TWELVE CONFIGURATIONS. QWEN2.5-0.5B-INSTRUCT, 256 GENERATED TOKENS, 20 MEASUREMENTS EACH (4 PROMPTS × 5 REPEATS). ENERGY IS RAPL PACKAGE−0, NET OF AN 11.23 W IDLE BASELINE.

| Precision | Placement | tok/s | J/token | Gross W | Net W | Perplexity | Needle |
|---|---|---|---|---|---|---|---|
| bf16 | P-1t | 16.1 | **0.1758** | 14.00 | 2.77 | 12.427 | 3/3 |
| bf16 | P-smt | 15.1 | 0.1878 | 14.01 | 2.77 | 12.427 | 3/3 |
| bf16 | E | 14.4 | 0.1987 | 14.01 | 2.78 | 12.427 | 3/3 |
| bf16 | LP-E | 7.0 | 0.4053 | 14.00 | 2.77 | 12.427 | 3/3 |
| fp32 | P-1t | 12.0 | 0.2313 | 14.00 | 2.77 | 12.449 | 3/3 |
| fp32 | E | 11.8 | 0.2357 | 14.00 | 2.77 | 12.449 | 3/3 |
| fp32 | P-smt | 11.1 | 0.2515 | 14.00 | 2.77 | 12.449 | 3/3 |
| fp32 | LP-E | 7.0 | 0.4019 | 14.00 | 2.77 | 12.449 | 3/3 |
| int8 | P-1t | 27.7 | 0.1281 | 14.78 | 3.55 | 109.953 | 0/3 |
| int8 | P-smt | 22.6 | 0.1470 | 14.59 | 3.36 | 109.522 | 0/3 |
| int8 | E | 22.5 | 0.1998 | 15.70 | 4.47 | 110.608 | 0/3 |
| int8 | LP-E | 11.2 | 0.3209 | 14.82 | 3.59 | 106.391 | 0/3 |

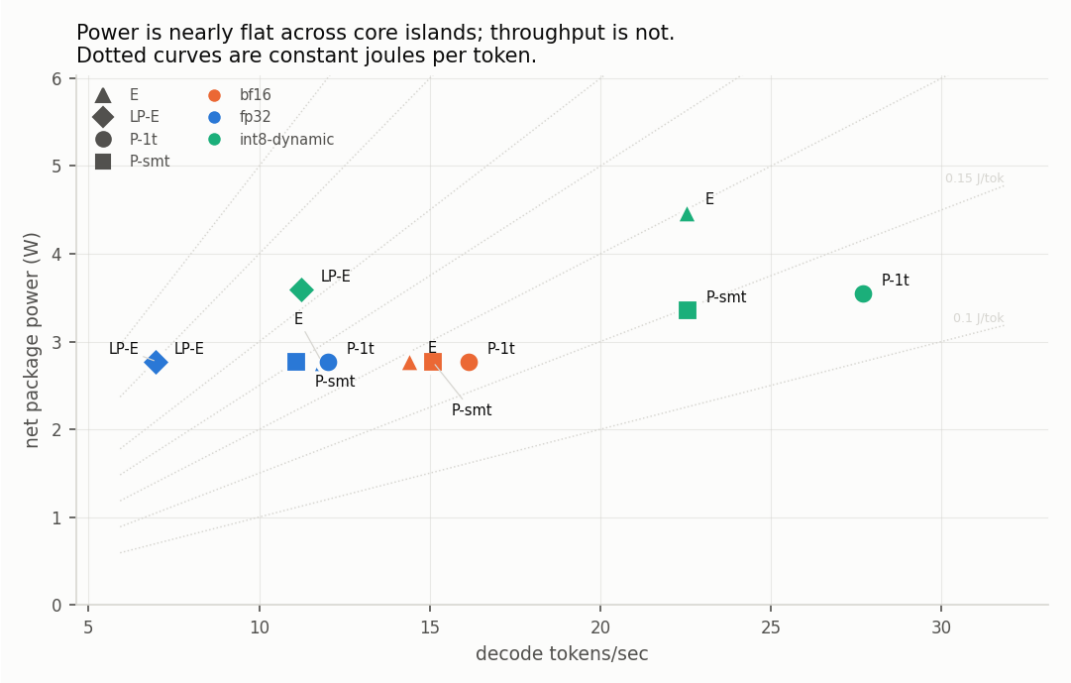



Fig. 1. Net package power against decode throughput, one point per configuration. fp32 (blue) and bf16 (orange) lie on a horizontal line at 2.77 W while throughput varies more than twofold; only int8 (green) departs from it. Dotted curves are constant joules per token, so a point's distance along a curve is pure duration. This is the whole argument of Section IV-A in one panel.

`Q4_K_M`; 1.876 against 1.884 for `Q8_0`). The flatness is therefore a property of the platform under this workload rather than of one inference framework.

The island penalty itself is smaller under llama.cpp (1.66–1.88× against 2.306× for bf16 under PyTorch), which is expected: the ratio depends on how completely an engine exploits the performance cores, and a faster engine widens the gap it must then be penalised against. The direction and the mechanism are unchanged; only the magnitude is engine-dependent, and we report it as such.

### C. Precision, and its interaction with placement

On the performance cores, fp32 costs 1.319× [1.309, 1.330] the energy of bf16 at 0.745× the throughput, with perplexity marginally worse (12.449 against 12.427). bf16 is better on every axis measured.

That advantage is not uniform across islands:

| Island | fp32/bf16 energy | fp32/bf16 tok/s |
|---|---|---|
| P-1t | 1.319 [1.309, 1.330] | 0.745 |
| P-smt | 1.335 [1.303, 1.367] | 0.732 |
| E | 1.173 [1.132, 1.205] | 0.833 |
| LP-E | **0.994 [0.989, 1.000]** | 0.991 |

**On the low-power island the bf16 advantage vanishes entirely.** fp32 and bf16 reach the same 7.0 tok/s and the same ≈0.40 J/token. The likely explanation is that the LP-E cores lack the bf16 instruction support available on the performance cores, so bf16 executes at fp32 rates; confirming this requires instruction-level profiling our harness does not perform, and

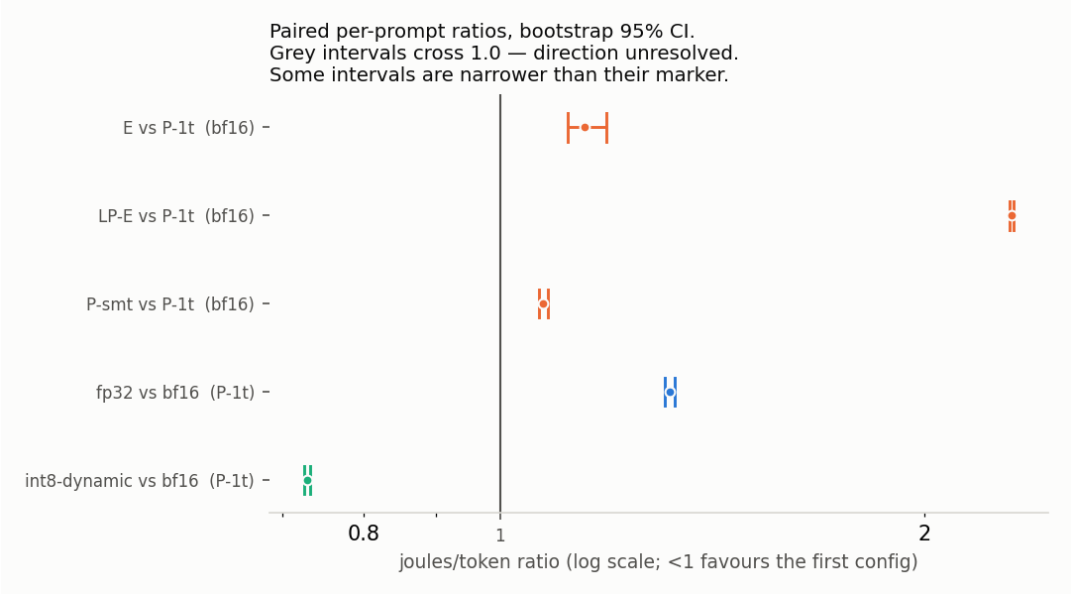



Fig. 2. Paired per-prompt energy ratios with bootstrap 95% intervals, log scale, rule at unity. Every comparison in this study on one axis. Several intervals are narrower than their marker.

TABLE II
ENERGY RATIO AGAINST INVERSE THROUGHPUT RATIO, BOTH MEASURED. FOR THE MEMORY-BOUND PRECISIONS THE TWO AGREE WITHIN 1.1%. INT8, THE ONE PRECISION WHOSE POWER IS NOT FLAT, IS ALSO THE ONE WHERE THE IDENTITY FAILS.

| Precision | Placement | Energy | 1/tput | Diff. |
|---|---|---|---|---|
| fp32 | P-smt | 1.086 | 1.088 | −0.16% |
| fp32 | E | 1.021 | 1.017 | +0.42% |
| fp32 | LP-E | 1.737 | 1.727 | +0.62% |
| bf16 | P-smt | 1.073 | 1.069 | +0.42% |
| bf16 | E | 1.148 | 1.136 | +1.10% |
| bf16 | LP-E | 2.306 | 2.296 | +0.43% |
| int8 | P-smt | 1.142 | 1.220 | −6.39% |
| int8 | E | 1.558 | 1.222 | **+27.51%** |
| int8 | LP-E | 2.509 | 2.447 | +2.54% |

we record it as a hypothesis.

The consequence for scheduling is concrete: **precision and placement cannot be chosen independently**. A runtime that migrates a bf16 workload to the low-power island does not merely slow it by that island's ratio—it also forfeits the entire benefit of the precision it was running.

### D. Per-tensor int8 saves energy and is still the wrong choice

int8-dynamic on the performance cores costs 0.730× [0.726, 0.733] the energy per token of bf16: a 27% saving, with the interval far from unity. The saving is nonetheless unusable. Perplexity rises from 12.427 to 109.953, an increase of 785%, and needle retrieval falls from 3/3 to 0/3.

The result is therefore a cost–quality trade rather than an absence of benefit. The question it raises is whether eight bits is simply too few, or whether the fault lies in how the scale is assigned. We test this directly.

TABLE III
QUANTIZATION QUALITY AT MATCHED BIT WIDTH. `Q8_0` AND INT8-DYNAMIC ARE BOTH 8-BIT AND DIFFER ONLY IN WHETHER THE SCALE IS ASSIGNED PER BLOCK OR PER TENSOR. PERPLEXITY OVER THE SAME CORPUS AND TOKENIZER IN EVERY ROW.

| Scheme | Engine | Bits | Granularity | Δ perplexity |
|---|---|---|---|---|
| bf16 | PyTorch | 16 | — | — |
| `Q8_0` | llama.cpp | 8 | per-block | +4.7% |
| int8-dynamic | PyTorch | 8 | per-tensor | **+785%** |
| `Q4_K_M` | llama.cpp | 4 | per-block | +7.4% |

### E. Granularity, not bit width

Comparing a per-tensor 8-bit scheme against a per-block 4-bit scheme would vary two things at once. We therefore add `Q8_0`, a GGUF per-block scheme at *the same eight bits* as PyTorch's dynamic quantization, served through llama.cpp. Bit width is then held constant and only granularity differs (Table III).

At eight bits, per-block scaling costs 4.7% perplexity and per-tensor scaling costs 785%: a factor of 167 in quality damage attributable to granularity alone, with bit width fixed. Halving the bit width to four, while keeping per-block scales, costs 7.4%—still two orders of magnitude less than the 8-bit per-tensor scheme. Bit width is not the operative variable on this workload; granularity is. The finding is consistent with the quantization literature [14], [15], [16], [18], [19], and we contribute the energy accounting rather than the scheme.

Two caveats bound what this comparison supports. The per-block schemes are served by a different runtime, so any *energy* difference between engines confounds quantization with implementation and we do not report one as a finding; the perplexity comparison is unaffected, since every row is scored over the same corpus with the same tokenizer. And a quality probe this thin (Section III) establishes that a scheme

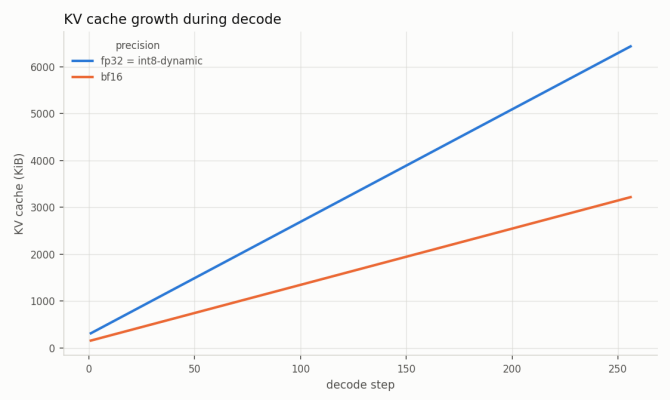



Fig. 3. KV cache size at every decode step. fp32 and int8-dynamic are byte-identical because dynamic quantization touches only Linear weights; bf16 is exactly half. Quantizing weights does not help on the axis that dominates long-context footprint.

has not collapsed, not that it is equivalent to full precision.

### F. Quantization does not shrink the KV cache

Dynamic quantization touches only Linear weights, so the KV cache remains at the unquantized dtype. The measured per-step curves for fp32 and int8 are byte-identical, both reaching 6,408 KiB after 256 decode steps, while bf16 reaches exactly half at 3,204 KiB (Fig. 3). For long-context work, where the cache rather than the weights dominates footprint, per-tensor int8 therefore offers no relief on the axis that matters while bf16 offers a factor of two.

### G. Placement changes the arithmetic, not the output

Greedy decode produced byte-identical generated text on all four islands, for every precision and prompt, verified by digest. Placement affects cost, not tokens.

The arithmetic beneath is not identical. int8 perplexity varies by placement across a 4% range—106.391 on LP-E against 110.608 on E—while fp32 and bf16 perplexity is constant to three decimal places. Different islands dispatch quantized kernels differently and the numerics differ measurably; greedy argmax is simply robust to a difference of that size.

Three consequences, in increasing order of practical weight. Any claim of bit-exact reproducibility across islands is false for quantized models, which matters for regression testing and for auditing a deployed system. A sampling-based decoder has no argmax margin to absorb the difference, so the invariance observed here does not transfer to it. And a scheduler that migrates a quantized workload between islands mid-generation is changing the arithmetic underneath a partially-built KV cache: our measurements cannot say whether that is harmless, because we never migrate mid-generation, and we flag it as a hazard that heterogeneous-aware runtimes should establish before relying on free migration.

### H. Prefill behaves oppositely, and confirms the mechanism

The argument of this paper rests on decode being memory-bound while prefill is not (Section II-B). Because the harness

TABLE IV
THROUGHPUT RATIOS AGAINST P-1T, BF16, PAIRED PER-PROMPT. SMT HELPS THE COMPUTE-BOUND PHASE AND HURTS THE MEMORY-BOUND ONE; THE LOW-POWER ISLAND IS RELATIVELY WORSE AT PREFILL, WHICH IS THE COMPUTE-BOUND PHASE.

| Placement | Prefill | Decode |
|---|---|---|
| P-smt | **1.193 [1.102, 1.370]** | **0.936 [0.926, 0.943]** |
| E | 0.933 [0.809, 1.012] | 0.881 [0.860, 0.901] |
| LP-E | 0.370 [0.367, 0.374] | 0.435 [0.434, 0.437] |

TABLE V
THE ISLAND PENALTY AGAINST MODEL SIZE, BF16, PAIRED PER-PROMPT. THE IDENTITY OF SECTION IV-A HOLDS ON BOTH MODELS.

| Model | LP-E vs. P-1t energy | Identity error |
|---|---|---|
| Qwen2.5-0.5B | 2.306 [2.301, 2.314] | +0.43% |
| SmolLM2-135M | **3.712 [3.159, 4.342]** | +0.68% |

times the two phases separately, that premise can be tested rather than assumed, on the same hardware in the same runs.

Simultaneous multithreading provides the test. Its effect reverses between the phases (Table IV): enabling SMT makes prefill $1.193\times$ faster and decode $0.936\times$ slower, with both intervals excluding unity. This is precisely the signature the two-regime account predicts. A compute-bound phase leaves issue slots idle that a sibling thread can fill, so SMT helps; a memory-bound phase is already waiting on DRAM, so the sibling adds cache contention and no useful parallelism.

The low-power island shows the same asymmetry from the other direction. It reaches $0.435\times$ the performance island's decode throughput but only $0.370\times$ its prefill throughput: relatively worse at the phase that demands compute. The E-core prefill ratio is the one comparison in this table whose interval spans unity, and we draw no conclusion from it.

Prefill nonetheless contributes little to the energy totals reported here, accounting for 2.3% of the measurement window on P-1t and 2.7% on LP-E at 256 generated tokens. That is why the rest of this paper reports decode: at realistic generation lengths it is almost the entire cost. The phases are separated so that this could be established rather than assumed, and so that the premise underlying the mechanism could be tested independently.

### I. Generalisation: model scale and context length

Two further experiments bound the result. Both use the protocol of Section III without modification.

*a) Model scale:* Repeating the full matrix on SmolLM2-135M, a model roughly a quarter the size, does not weaken the effect—it strengthens it (Table VI). The penalty for placing decode on the low-power island is 61% larger on the smaller model, and the duration identity holds on both.

The mechanism is visible in the throughput. The performance cores exploit the smaller model—16.1 to 34.7 tok/s, a $2.2\times$ gain—while the low-power island barely moves, 7.0 to 8.3 tok/s, a $1.2\times$ gain. Less weight traffic makes the workload relatively less memory-bound, so core capability begins to matter, and that is precisely what the low-power island does not supply.

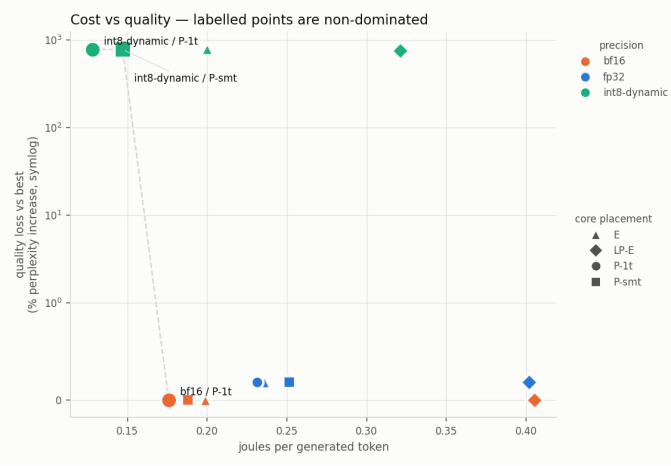



Fig. 4. Cost against quality loss, frontier dashed. Two of the three non-dominated configurations are int8 variants that score 0/3 on needle retrieval: cheaper, and unusable.

The same mechanism erases the quantization benefit. On SmolLM2, int8 costs $1.053\times$ [1.007, 1.101] the energy of bf16—marginally *more* expensive, with the interval excluding unity on the opposite side to the $0.730\times$ measured on Qwen. Quantization saves energy only to the degree the model is memory-bound; perplexity still rises from 13.4 to 40.8. We note that the needle probe carries no signal at this model size, SmolLM2-135M being a base model rather than instruction-tuned, and do not report it as a quality comparison there.

Package power was again flat: medians of 14.00 W for fp32 and bf16 and 14.26 W for int8, on a different model and against a different idle baseline (9.66 W). This is the third independent observation of the invariance.

*b) Context length:* Generating 1024 tokens rather than 256 quadruples the KV cache, from 3,204 to 12,384 KiB, and costs both islands about 12% throughput as attention over a longer context grows more expensive. The ratio is essentially unmoved: $2.251\times$ against $2.306\times$, with package power still exactly 14.00 W on both islands at four times the context length.

We report this comparison without an interval. It was run with two prompts rather than four, leaving two paired ratios—below the three our bootstrap procedure requires before it will report one (Section III-C). The point estimate is sound and the conclusion is stable across the two lengths, but we flag the limitation rather than present a number the design cannot support.

### J. The Pareto frontier is necessary, not sufficient

The frontier contains `bf16/P-1t`, `int8/P-1t` and `int8/P-smt`. Two of the three score 0/3 on needle retrieval and carry a 785% perplexity penalty (Fig. 4): non-dominated because genuinely cheaper, and genuinely unusable.

This is a limitation of the method rather than a result. A frontier identifies what is not dominated, not what is worth choosing, and must be read alongside the quality column rather than instead of it. The actionable output of this study is bf16 on the performance cores, one thread per core.

## V. Discussion

### A. Implications for schedulers

The practical recommendation is the opposite of the one the island naming suggests: on this class of part, sustained LLM decode should not be migrated to the efficiency or low-power islands in pursuit of energy savings. Doing so costs $1.148\times$ and $2.306\times$ the energy per token respectively, and returns nothing, because there is no power reduction to collect.

The generalisable form of the recommendation is not "use the big cores". It is that the question is settled by one measurement: whether package power varies across islands for the workload in question. Where it does, a power–duration trade exists and migration can win. Where it does not—as here, because decode is memory-bound—migration only buys duration, and duration is the entire cost.

A second consequence is that a runtime cannot treat precision and placement as independent knobs. Migrating a bf16 workload to the low-power island forfeits the bf16 advantage entirely (Section IV-C), so the cost of migration is larger than the island ratio alone implies.

### B. Implications for practitioners

For deployment on this class of hardware: bf16 on the performance cores, one thread per core. SMT costs throughput at every precision and on both models measured—6–18% on Qwen2.5-0.5B and 16–23% on SmolLM2-135M—which is the expected behaviour for a memory-bound workload whose sibling thread adds contention without adding useful parallelism.

Throughput should not be used as a proxy for energy without first checking that power is flat. Where it is flat the proxy is exact, and where it is not—int8 here—it misleads by up to 27%.

Finally, quantization should be evaluated at the level of scale granularity rather than bit width. The scheme measured here is fast, genuinely cheaper in energy, and unusable.

### C. What the difference is worth

Work on the sustainability of machine learning has concentrated on training [20], [21], on reporting standards [22], and on the argument that efficiency should count alongside accuracy as a research objective [23]. Inference at the edge has received less attention, and the quantity this paper measures is small per event and large only in aggregate, which is the characteristic shape of an efficiency result worth acting on. It is worth stating the magnitude explicitly rather than leaving the reader to assume it.

The avoidable energy is the gap between the best and worst placement at fixed precision and byte-identical output: $0.4053 - 0.1758 = 0.2295$ J per generated token (Table I). A device generating $10^4$ tokens per day—on the order of twenty to thirty short interactions, and the weakest assumption in what follows—therefore wastes 838 kJ, or 0.233 kWh, over a year by scheduling decode on the low-power island. At the 2025 global average grid intensity of 458 $gCO_2e$/kWh [24] that is 107 $gCO_2e$ per device per year; the IEA's $CO_2$-only figure of 435 g/kWh [25] gives 101 g.

TABLE VI
AVOIDABLE ENERGY FROM PLACEMENT, SCALED. THE 2.306× RATIO IS MEASURED; EVERYTHING ELSE IS ARITHMETIC OVER STATED ASSUMPTIONS. THE ESTIMATE IS LINEAR IN THE TOKEN RATE, WHICH IS THE LEAST CERTAIN INPUT.

| Tokens/device/day | kWh/device/yr | GWh/yr | $tCO_2e$/yr |
|---|---|---|---|
| 1 000 | 0.023 | 1.8 | 830 |
| 5 000 | 0.116 | 9.1 | 4 150 |
| 10 000 | 0.233 | 18.1 | 8 290 |
| 50 000 | 1.163 | 90.5 | 41 460 |

Across a population the arithmetic is linear. Gartner places 2025 AI-PC shipments at 77.8 million units [26]. One cohort of that size, under the same assumption, accounts for **18.1 GWh and roughly 8 300 tonnes $CO_2$e per year** attributable to a single scheduling decision. Table VI gives the sensitivity to the token rate, which dominates the estimate.

Three qualifications keep this honest. The figures assume every device makes the worse choice, so they bound the opportunity rather than describe current waste. They cover only the cohort shipped in a single year, and say nothing about the installed base. And they are linear in a token rate we have not measured and that no public figure establishes well.

What makes the quantity interesting is not its size but its cost. It requires no new hardware, no accuracy sacrifice and no change to the model: the output is byte-identical across placements (Section IV-G). It is recovered by scheduling, which is among the cheapest interventions available in a field whose efficiency gains usually demand retraining, quantization or silicon. Against the energy of training the models being served it is negligible; against the cost of obtaining it, it is nearly free.

## VI. THREATS TO VALIDITY

*One machine, one SoC generation.* Nothing here generalises beyond Meteor Lake without further measurement. The result is explicitly a property of a platform whose package power is flat across islands for this workload; a part on which that does not hold would behave differently, and Section VII-A discusses one that apparently does not.

*Model scale.* Models below 1 B parameters, bounded by available memory. Section IV-I shows the ordering holds across a $4\times$ range of model size and that the penalty grows as models shrink, but 7 B—where memory pressure is far higher and package power may cease to be flat—remains untested and is the most likely place these conclusions break.

*Idle baseline.* Net energy is gross minus idle power times duration, and the baseline is the least trustworthy term in that expression. Here it is 11.23 W against a gross draw of 14.00 W, so it accounts for roughly 80% of each reading and the net signal is the remaining 2.77 W. This is the strongest argument for reporting ratios rather than absolute joules: if two configurations draw equal power, their net energy ratio equals their duration ratio whatever baseline is subtracted, so every comparison in Section IV is baseline-independent while the absolute figures in Table I are not. An earlier iteration of this study measured 24-token generations, where the baseline exceeded the net signal outright; we report that dataset and its failures alongside this one. We also report a failed attempt at a per-configuration baseline which made the data substantially noisier, and the reason: a baseline sampled immediately after a burst of work captures the CPU decaying out of turbo rather than idling.

*Batch size one.* Every measurement is unbatched. Batching amortises each weight read across several sequences, raising arithmetic intensity and moving decode back toward the compute-bound regime, so the power invariance reported here should not be assumed to survive it. The single-request case is the on-device regime this paper addresses; batched serving is a different setting and an obvious next measurement.

*Package domain includes the iGPU.* CPU and iGPU energy cannot be fully separated within a run on an integrated part.

*No external instrument.* We did not validate RAPL against a wall meter or a shunt on this part. We rely on published characterisation of RAPL against full-system power [3] rather than on a measurement of our own, which is a dependency worth naming. It bears mainly on absolute joules; the comparisons in Section IV are ratios taken from the same counter under identical conditions, where a systematic instrument offset cancels.

*Quality measure.* Perplexity and a single needle probe are thin, and the needle probe carries no signal on a base model: SmolLM2-135M fails it at every precision, so it cannot discriminate there and we do not use it to. It was adequate for its purpose, which was detecting the int8 collapse on the instruction-tuned model, and nothing more.

*No hardware performance counters.* The SMT penalty and the bf16/LP-E equivalence are both inferred from throughput. Cache-miss and stall-cycle counters would convert both from plausible accounts into measurements, and the missing-bf16-support explanation in Section IV-C in particular remains a hypothesis.

*Machine quiescence.* The host was a laptop in ordinary use and some trials show excursions consistent with background load. This widens intervals rather than biasing them in a particular direction, so the effects reported here are if anything understated.

## VII. RELATED WORK

### A. A contradicting result on ARM mobile, and why both can hold

The conclusion reached here is the opposite of the one reached by MNN-AECS [6], which reduces LLM decoding energy on mobile devices precisely by *dynamically selecting low-power CPU cores*, reporting a 23% energy reduction across five Android and two iOS devices. Our measurement says that placing decode on the low-power island costs $2.306\times$ the energy per token. The disagreement is not incidental and we do not attempt to explain it away; it is the most important thing to settle about this result.

We believe both findings are correct on their own platforms, and that the difference is a property of the hardware rather than of the methodology. Three distinctions matter.

*The islands are not analogous.* ARM big.LITTLE pairs out-of-order performance cores with genuinely low-power in-order cores designed to run at a small fraction of the big cores' power. Intel's E-cores are throughput-oriented rather than low-power-oriented: they exist to deliver area-efficient multithreaded throughput, not to sip power. Only the LP-E island is positioned as a low-power domain, and it contains two cores. The name "efficiency core" spans two quite different design intents across the two ecosystems.

*The power floor differs.* Our central measurement is that net package power is approximately invariant across islands (Section IV-A). On a part where the low-power island does achieve a substantially lower operating power, the trade that MNN-AECS exploits is available and ours is not. The question is empirical for each platform, and the answer we report is specific to the one we measured.

*The objective differs.* MNN-AECS optimises energy subject to an accepted slowdown threshold. Where power is flat, no such trade exists: slowing down costs energy proportionally and buys nothing.

The contribution of this paper is therefore narrower and, we think, more useful than "efficiency cores do not save energy". It is that the mobile result does not transfer to x86 heterogeneous SoCs, that the reason is measurable rather than conjectural, and that the measurement needed to tell the two cases apart is a single one: whether package power varies across islands for the workload in question.

### B. Energy of LLM inference

Most published energy figures for LLM inference describe datacenter GPUs. Samsi et al. [1] benchmark multi-node, multi-GPU inference at scale, and Luccioni et al. [2] compare the deployment cost of model categories in energy and carbon per thousand inferences, establishing that generative architectures are orders of magnitude more expensive than task-specific ones. Both measure at the scale and on the hardware where most inference tokens are currently produced.

Characterisation at the edge is more recent and more fragmented. Zhang and Huang [27] report cases in which CPUs outperform integrated GPUs for on-device inference. Taherin et al. [28] characterise inference across edge SoC generations, backends and quantization levels, and Zhang et al. [29] show that independently operating CPU, GPU and memory frequency governors leave substantial latency on the table at equal energy. What this body of work does not yet provide, and what we contribute, is a ground-truth energy comparison across the *core islands of a single x86 SoC* at fixed output, with the power and duration terms reported separately rather than multiplied together.

### C. Why decode is memory-bound

The separation of prefill and decode into a compute-bound and a memory-bound phase is established by roofline analysis of transformer inference [9]: prefill reuses weights across the whole prompt and sits near or beyond the ridge point, while single-token decode is a sequence of matrix–vector products at roughly one FLOP per byte. Serving systems are built around the consequence, managing memory rather than arithmetic as the scarce resource [10], and the major inference optimisations attack the same bottleneck from different directions: restructuring attention to reduce memory traffic [30], or amortising the per-token weight read across several candidate tokens [31]. All three presuppose that decode is memory-bound; we measure a consequence of that premise rather than assuming it.

We rely on this property for the mechanism in Section IV-A: if the bottleneck is off-core, the identity of the cores issuing the loads has little influence on the power drawn, and the energy a placement costs is set by how long it occupies the memory system. To our knowledge this consequence has not previously been measured directly as package power invariance across heterogeneous islands.

### D. Race to idle

Whether to run fast and sleep or to run slowly and stay busy is a long-settled question in the abstract and a workload-dependent one in practice. Le Sueur and Heiser [4] examine the decline of DVFS effectiveness as static power grows and idle states improve; Albers and Antoniadis [5] give the theoretical treatment of speed scaling with a sleep state; Awan and Petters [32] develop race-to-halt strategies for embedded real-time systems; and Kim et al. [33] analyse racing and pacing heuristics both theoretically and empirically, finding that neither dominates universally.

The usual framing of this debate presupposes a power–frequency trade: running slower is assumed to draw less power, and the question is whether the saving offsets the longer runtime. Our contribution to this line is a workload for which the premise fails outright. Across islands, running slower draws *the same* power (Section IV-A), so there is no trade to evaluate and racing wins by default. We regard this as a sharper case than the ones usually studied rather than a new heuristic.

### E. Heterogeneous and asymmetric multicore energy

The intuition that migrating work to smaller cores saves energy was developed on ARM big.LITTLE and related asymmetric designs. Catalán et al. [11] optimise matrix multiplication across asymmetric cores for performance and energy jointly; Butko et al. [12] explore the trade-off space across heterogeneous configurations; and Coutinho et al. [13] measure performance and energy for parallel applications on heterogeneous multi-processing systems. This literature establishes both that the trade is real on those parts and that it is workload-dependent.

Two things make the present setting different. The workload is sustained and memory-bound rather than bursty and compute-bound, which is the regime in which the premise of the trade is most likely to fail. And the hardware divides differently: Intel's E-cores are throughput-oriented rather than low-power-oriented (Section II), so a result established on big.LITTLE does not transfer on the strength of the shared label alone. Section VII-A works through the one case where this matters most.

### F. Quantization granularity

Post-training quantization of LLM weights is a mature area, and its consistent lesson is that the assignment of scales matters at least as much as the remaining bit width. LLM.int8() [14] isolates outlier features for higher-precision treatment; GPTQ [15] performs second-order-informed per-layer rounding; AWQ [16] scales channels by activation magnitude and targets on-device deployment specifically; SmoothQuant [18] migrates activation outliers into the weights so that both can be quantized. All preserve quality at bit widths where uniform per-tensor schemes fail, and the broader survey [19] frames granularity as a primary design axis rather than an implementation detail.

The scheme we measure sits at the naive end of that spectrum deliberately. Our contribution here is not a new quantization method but an energy accounting of one: we show that the per-tensor scheme is genuinely cheaper in energy, by 27%, and that the quality cost makes the saving unusable—so the relevant design axis for energy-aware deployment is granularity, not bit width.

### G. RAPL as an instrument

Khan et al. [3] characterise RAPL against full-system power on production hardware, reporting close agreement, negligible overhead and microsecond-scale counter updates. The counters' unprivileged accessibility was subsequently removed: PLATYPUS [7] recovers AES-NI keys from SGX enclaves and defeats KASLR through RAPL alone, and related work extracts further secrets through RAPL-induced side channels [8]. Any reproduction of measurements like ours therefore requires a deliberate relaxation of permissions, which we document in the artifact rather than leave implicit.

## VIII. Conclusion

On a heterogeneous consumer SoC, the island named for efficiency is not the efficient choice for LLM decode. The low-power island costs $2.306\times$ the energy per generated token of the performance island, and it does so for a reason that is measurable rather than conjectural: package power is flat at 14.00 W across islands, so energy per token is duration, and a slower island is simply a more expensive one. The measured energy ratios match inverse throughput to within 1.1% wherever power is flat, and depart from it—by up to 27.5%—for the one precision where power is not.

The underlying cause is that decode is memory-bandwidth-bound, which makes the result portable as a method even where it is not portable as a number. A platform whose low-power island does achieve materially lower power under this workload would admit the trade our platform denies, and such a result has been reported on ARM mobile. The two findings are reconciled by a single measurement, which we recommend as the first thing to check before scheduling inference across heterogeneous cores: does package power vary across islands for this workload? If it does not, there is nothing to trade, and the right policy is to race to idle.

## Acknowledgment

The author used Anthropic's Claude (Claude Opus 5) as a writing and engineering assistant. It contributed to drafting the manuscript text, to the statistical analysis code—paired per-prompt ratios, bootstrap confidence intervals—and figure generation in `analyze.py`, and to corrections of the measurement harness. The research questions, experimental design, all measurements, and the interpretation of results are the author's own. The author takes full responsibility for the content.

## Artifact Availability

The measurement harness, the complete dataset including every discarded run, and the analysis that generates every table and figure in this paper are archived under a permanent DOI, withheld for double-anonymized review because the repository identifies the author and supplied on the title page. Five superseded runs are included deliberately, each with a written account of why it was rejected—short measurement windows, prompts cycled by repeat index, an unsettled warmup, a per-configuration baseline that made the data noisier, and a half-measured dataset—because the reason a measurement was discarded is part of the method.

## References

[1] S. Samsi, D. Zhao, J. McDonald, B. Li, A. Michaleas, M. Jones, W. Bergeron, J. Kepner, D. Tiwari, and V. Gadepally, "From words to watts: Benchmarking the energy costs of large language model inference," in *2023 IEEE High Performance Extreme Computing Conference (HPEC)*. IEEE, 2023.

[2] A. S. Luccioni, Y. Jernite, and E. Strubell, "Power hungry processing: Watts driving the cost of AI deployment?" in *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2024, pp. 85–99.

[3] K. N. Khan, M. Hirki, T. Niemi, J. K. Nurminen, and Z. Ou, "RAPL in action: Experiences in using RAPL for power measurements," *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, vol. 3, no. 2, pp. 1–26, 2018.

[4] E. Le Sueur and G. Heiser, "Slow down or sleep, that is the question," in *USENIX Annual Technical Conference (ATC)*, 2011.

[5] S. Albers and A. Antoniadis, "Race to idle: New algorithms for speed scaling with a sleep state," *ACM Transactions on Algorithms*, vol. 10, no. 2, pp. 1–31, 2014.

[6] Z. Huang, C. Niu, Z. Wang, J. Xue, H. Zhang, Y. Wang, Z. Xin, X. Jiang, C. Lv, F. Wu, and G. Chen, "MNN-AECS: Energy optimization for LLM decoding on mobile devices via adaptive core selection," *arXiv preprint*, 2025.

[7] M. Lipp, A. Kogler, D. Oswald, M. Schwarz, C. Easdon, C. Canella, and D. Gruss, "PLATYPUS: Software-based power side-channel attacks on x86," in *2021 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2021, pp. 355–371.

[8] Z. Zhang, S. Liang, F. Yao, and X. Gao, "Red alert for power leakage: Exploiting Intel RAPL-induced side channels," in *Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security (ASIA CCS)*, 2021, pp. 162–175.

[9] Z. Yuan, Y. Shang, Y. Zhou, Z. Dong, Z. Zhou, C. Xue, B. Wu, Z. Li, Q. Gu, Y. J. Lee, Y. Yan, B. Chen, G. Sun, and K. Keutzer, "LLM inference unveiled: Survey and roofline model insights," *arXiv preprint*, 2024.

[10] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, "Efficient memory management for large language model serving with PagedAttention," in *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023, pp. 611–626.

[11] S. Catalán, F. D. Igual, R. Mayo, L. Piñuel, E. S. Quintana-Ortí, and R. Rodríguez-Sánchez, "Performance and energy optimization of matrix multiplication on asymmetric big.LITTLE processors," in *HiPEAC*, 2015.

[12] A. Butko, F. Bruguier, D. Novo, A. Gamatié, and G. Sassatelli, "Exploration of performance and energy trade-offs for heterogeneous multicore architectures," *arXiv preprint*, 2019.
[13] D. A. M. Coutinho, D. De Sensi, A. F. Lorenzon, K. Georgiou, J. Nunez-Yanez, K. Eder, and S. Xavier-de Souza, "Performance and energy trade-offs for parallel applications on heterogeneous multi-processing systems," *Energies*, vol. 13, no. 9, p. 2409, 2020.
[14] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, "LLM.int8(): 8-bit matrix multiplication for transformers at scale," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.
[15] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "GPTQ: Accurate post-training quantization for generative pre-trained transformers," in *International Conference on Learning Representations (ICLR)*, 2023.
[16] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, C. Gan, and S. Han, "AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 6, 2024.
[17] L. A. Barroso and U. Hölzle, "The case for energy-proportional computing," *Computer*, vol. 40, no. 12, pp. 33–37, 2007.
[18] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han, "SmoothQuant: Accurate and efficient post-training quantization for large language models," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023, pp. 38 087–38 099.
[19] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," in *Low-Power Computer Vision*. Chapman and Hall/CRC, 2022, pp. 291–326.
[20] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in NLP," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 3645–3650.
[21] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, "Carbon emissions and large neural network training," *arXiv preprint*, 2021.
[22] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of machine learning," *Journal of Machine Learning Research*, vol. 21, no. 248, pp. 1–43, 2020.
[23] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, 2020.
[24] Ember, "Global electricity review 2026," Ember, Tech. Rep., 2026, global average grid emissions intensity 458 $gCO_2e$/kWh in 2025, down from 533 in 2015. Accessed 2026-10-09.
[25] International Energy Agency, "Electricity 2026," IEA, Tech. Rep., 2026, global $CO_2$ intensity of electricity generation estimated at 435 g/kWh for 2025. Accessed 2026-10-09.
[26] Gartner, Inc., "Gartner says AI PCs will represent 31% of worldwide PC market by the end of 2025," Press release, Aug. 2025, projects 77.8 million AI PC shipments in 2025 and 143 million in 2026; a February 2026 update defers 50% penetration to 2028. Accessed 2026-10-09.
[27] H. Zhang and J. Huang, "Challenging GPU dominance: When CPUs outperform for on-device LLM inference," *arXiv preprint*, 2025.
[28] A. Taherin, S. T. Anvari, C. Amante, Y. Chen, R. Noroian, Z. Feric, N. B. Agostini, P. Zhao, J. Cano, B. Ren, Y. Wang, and D. Kaeli, "Hydra: Phase-aware workload characterization of LLM inference across edge SoC generations, backends, and quantization levels," *arXiv preprint*, 2026.
[29] Z. Zhang, P. Dash, Y. C. Hu, Q. Xu, J. Li, and H. Guan, "Dissecting the impact of mobile DVFS governors on LLM inference performance and energy efficiency," *arXiv preprint*, 2025.
[30] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "FlashAttention: Fast and memory-efficient exact attention with IO-awareness," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, pp. 16 344–16 359.
[31] Y. Leviathan, M. Kalman, and Y. Matias, "Fast inference from transformers via speculative decoding," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023, pp. 19 274–19 286.
[32] M. A. Awan and S. M. Petters, "Race-to-halt energy saving strategies," *Journal of Systems Architecture*, vol. 60, no. 10, pp. 796–815, 2014.
[33] D. H. K. Kim, C. Imes, and H. Hoffmann, "Racing and pacing to idle: Theoretical and empirical analysis of energy optimization heuristics," in *IEEE 3rd International Conference on Cyber-Physical Systems, Networks, and Applications (CPSNA)*, 2015, pp. 78–85.